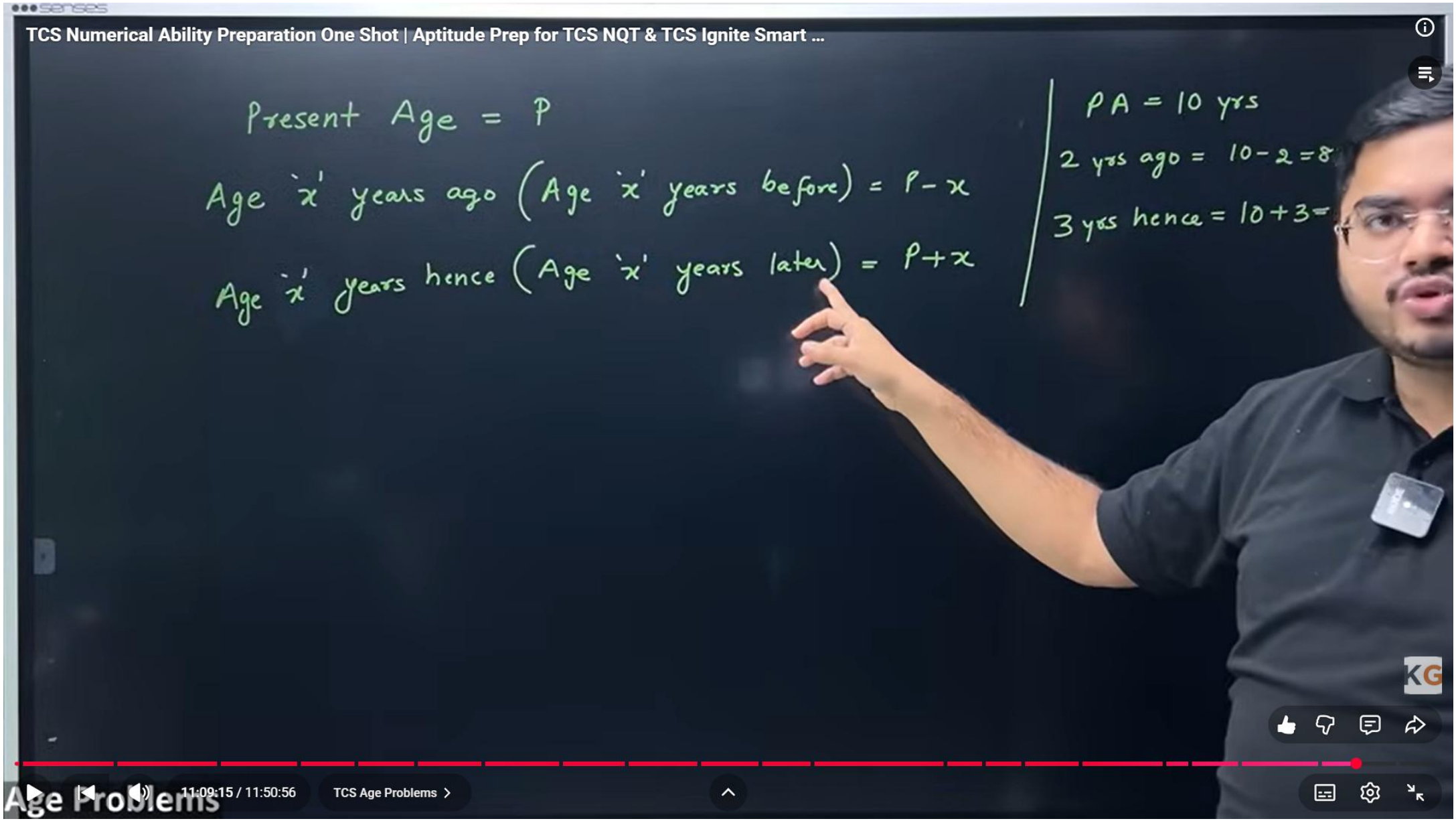
TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
Present Age = P
Age 'x' years ago (Age 'x' years before) = P − x
Age 'x' years hence (Age 'x' years later) = P + x
PA = 10 yrs
2 yrs ago = 10 − 2 = 8
3 yrs hence = 10 + 3 =
11:09:15 / 11:50:56
TCS Age Problems
Age Problems

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q1. The sum of ages of a father and a son is 90 years. 15 years ago, father's age was thrice the son's age. Find their present ages.

S 5

Father's PA = F

Son's PA = B

15 years ago

Father = F − 15

Son = B − 15

$F + B = 90$

$-F + 3B = 30$

$4B = 120$

$B = \frac{120}{4} = 30$

$F = 60$

$F - 15 = 3(B - 15)$

$F - 15 = 3B - 45$

$45 - 15 = 3B - F$

$3B - F = 30$

Age Problems

11:12:49 / 11:50:56

TCS Age Problems

senses

Q2. The sum of ages of a father and a son is 60 years. 10 years later, father's age was thrice the son's age. Find their present ages.

Father's PA = F

Son's PA = B

10 years later

Father = F+10

Son = B+10

$\boxed{F+B=60} \Rightarrow F=60-B$

$(F+10) = 3(B+10)$

$F+10 = 3B+30$

$(60-B)+10 = 3B+30$

$70-B = 3B+30$

$70-30 = 3B+B$

$40 = 4B$

$B = \frac{40}{4} = 10$

$B=10$

$F=50$

KG

senses

Q3: Three years ago, the age of father is 7 times as his son's age. At present the age of father is 5 times the age of his son then what is the sum of present ages of father and son?

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Father's PA = F
Son's PA = B

3 year ago

Father = F - 3
Son = B - 3

$F = 5B$

$(F-3) = 7(B-3)$

$F - 3 = 7B - 21$

$5B - 3 = 7B - 21$

$21 - 3 = 7B - 5B$

$18 = 2B$

$B = \frac{18}{2} = 9$

$F = 5B$
$= 5 \times 9$
$= 45$

$F = 45, B = 9$

$F + B = 45 + 9 = 54$

KG

senses

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Q3: Three years ago, the age of father is 7 times as his son's age. At present the age of father is 5 times the age of his son then what is the sum of present ages of father and son?

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$$\frac{3(7-1)}{7-5} = \frac{3(6)}{2} = \frac{18}{2} = 9$$

Present age of younger one

Son's PA = 9

Father's PA = 9 × 5 = 45

$$\text{PA (Younger)} = \frac{(\text{YEAR})[\text{TIMES} - 1]}{\text{TIMES} - \text{TIMES}}$$

KG

Age Problems 11:20:05 / 11:50:56 TCS Age Problems >

senses

Q4: At present, the age of father is 5 times the age of his son. After 3 years, the age of father will be 4 times the age of his son. Find the difference of present age of father and son?

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$$PA\ (\text{Younger}) = \frac{(\text{YEAR})[\text{TIMES}-1]}{\text{TIMES}-\text{TIMES}} = \frac{3\,(4-1)}{5-4} = \frac{3\times 3}{1} = 9$$

$\boxed{PA \text{ of Son} = 9}$ ✓

$\boxed{PA \text{ of Father} = 9\times 5 = 45}$ ✓

KG

senses

Q5. 10 years ago, Goli's mother is 4 times old as Goli. 10 years later, the mother will be twice as old as his son. What is the present age of Goli's Mother ?

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$$PA(\text{Younger}) = \frac{(\text{YEAR})[\text{TIMES}-1]}{\text{TIMES}-\text{TIMES}} = \frac{10(4-1)+10(2-1)}{4-2}$$

$$= \frac{(10\times3)+(10\times1)}{2} = \frac{30+10}{2} = \frac{40}{2} = 20$$

Goli's PA = 20

Goli's age 10 yrs ago = 20 − 10 = 10

Mother's age 10 yrs ago = 10 × 4 = 40

Mother's PA = 40 + 10 = 50

Goli's PA = 20

Goli's age 10 years later = 20 + 10 = 30

Mother's age 10 years later = 30 × 2 = 60

Mother's PA = 60 − 10 = 50

Age Problems

11:24:50 / 11:50:56 TCS Age Problems >

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Q6. Three years ago, the age of a mother was five times the age of her daughter. Two years hence, she will be thrice as old as her daughter. Find present age of daughter.

$$PA(\text{Younger}) = \frac{(\text{YEAR})[\text{TIMES}-1]}{\text{TIMES}-\text{TIMES}} = \frac{3(5-1)+2(3-1)}{5-3} = \frac{3(4)+2(2)}{2}$$

$$= \frac{12+4}{2} = 8$$

$\boxed{\text{Daughter PA} = 8}$

Daughter's age 3 years ago $= 8-3 = 5$

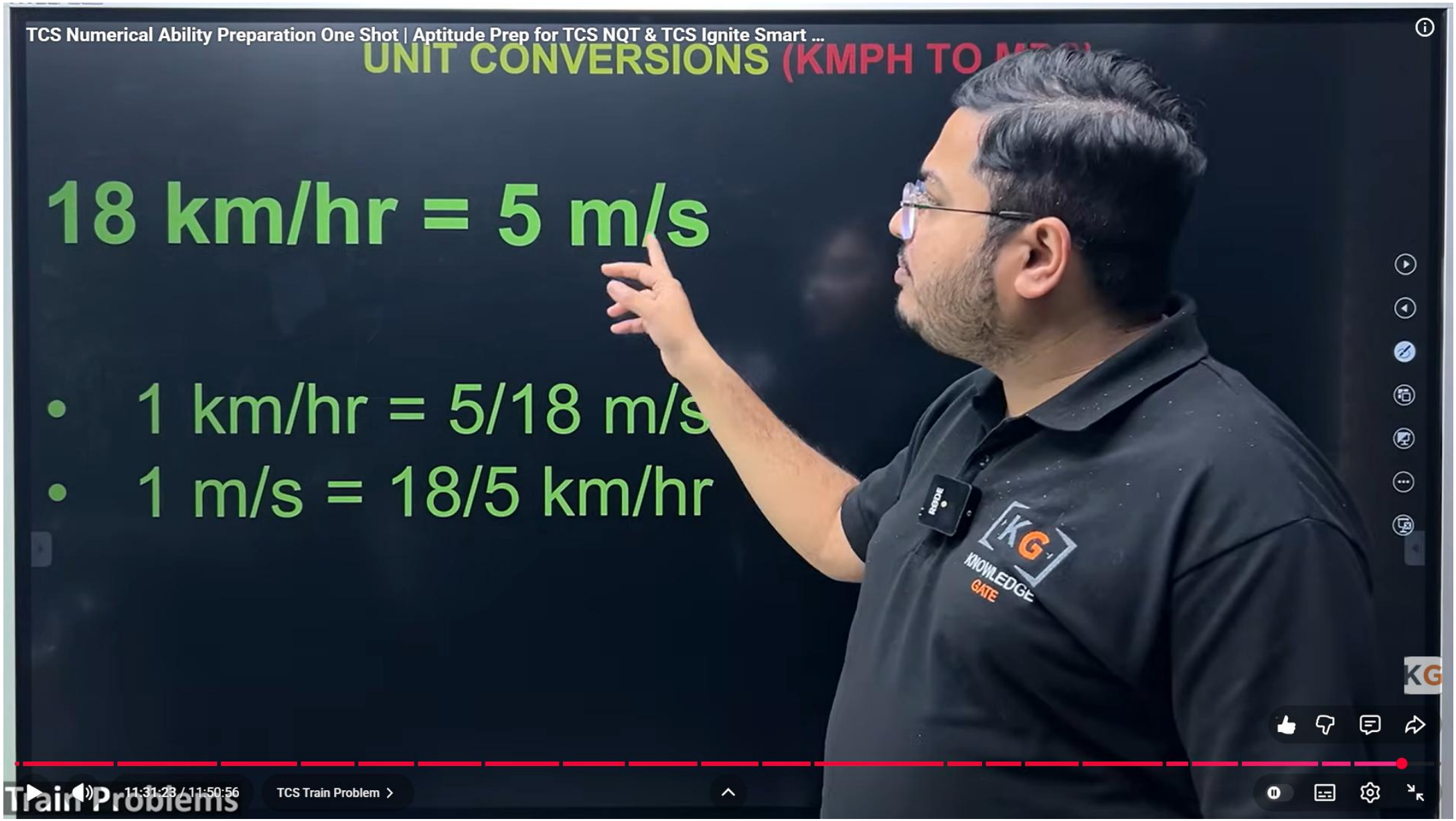
Mother's age 3 years ago $= 5 \times 5 = 25$

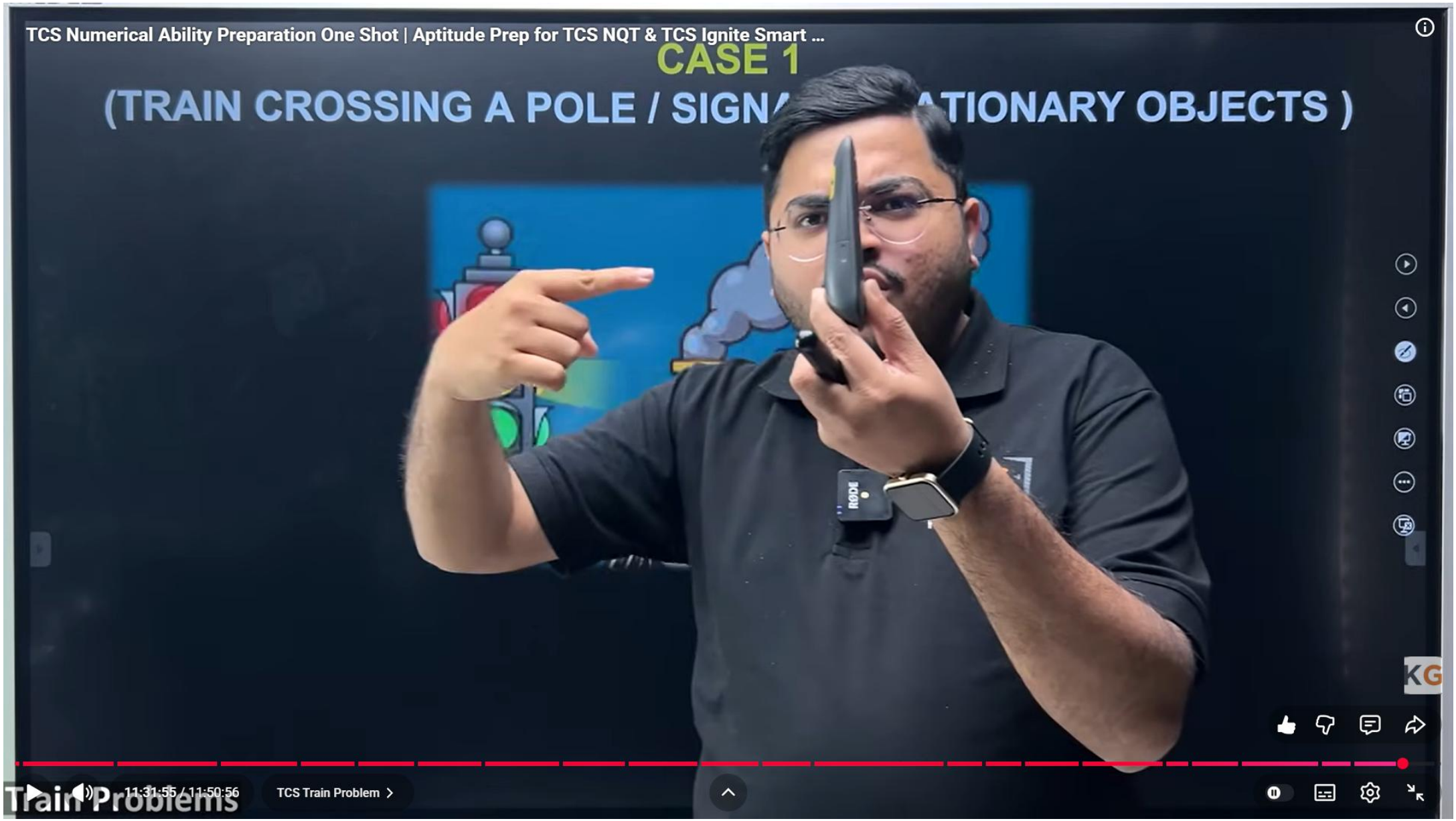
Mother's PA $= 25+3 = 28$

2 yrs later

10
30
28

Age Problems
11:28:41 / 11:50:56
TCS Age Problems


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
UNIT CONVERSIONS (KMPH TO
18 km/hr = 5 m/s
• 1 km/hr = 5/18 m/s
• 1 m/s = 18/5 km/hr
KNOWLEDGE GATE
11:31:23 / 11:50:56
TCS Train Problem
Train Problems


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
CASE 1
(TRAIN CROSSING A POLE / SIGN
TIONARY OBJECTS)
11:31:55 / 11:50:56
TCS Train Problem
Train Problems

Q1: Punjab Mail Express of length 80 meters goes to Delhi to visit Baba Ka Dhaba. Find the time taken (in seconds) by Punjab Mail Express to cross a pole if it is travelling with a speed of 80 km/hr?

length of train = 80

$$S = \frac{\text{length of train}}{t}$$

$$t = \frac{\text{length of train}}{S} = \frac{\cancel{80}}{\frac{5}{18} \times \cancel{80}} = \frac{18}{5} = \frac{36}{10} = 3.6 \text{ seconds}$$

18 km/hr = 5 m/s

1 km/hr = $\frac{5}{18}$ m/s

80 km/hr = $\frac{5}{18} \times 80$ m/s

Q2: Bhopal Shatabdi Express (length 150 meters) crosses Mathura Junction (110 meters long) with a speed of 90 km/hr. Find the time taken to cross the platform (in seconds).

$$S = \frac{\text{length of train + length of platform}}{t}$$

$$25 = \frac{150 + 110}{t}$$

$$t = \frac{260}{25} = \frac{1040}{100} = 10.4 \text{ seconds}$$

$\times 4$ $\times 4$

$$\begin{array}{r} 260 \\ \times\ 4 \\ \hline 1040 \end{array}$$

18 km/hr = 5 m/s

$\downarrow \times 5$ $\downarrow \times 5$

90 km/hr = 25 m/s

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CASE 3

TWO TRAINS CROSSING EACH OTHER

$\longrightarrow S_1$

$\longleftarrow S_2$

$$S = \frac{d}{t}$$

$$\text{Relative Speed } (S_1 + S_2) = \frac{\text{length of T1} + \text{length of T2}}{t_{cross}}$$

Train Problems
11:42:31 / 11:50:56
TCS Train Problem
KG

Q4: Two trains Jhelum Express (length 100 meters) and Kamayani Express (length 80 meters) are moving in opposite directions with speed of 60 km/hr and 54 km/hr. Find the time (in seconds) they take to cross each other completely?

$$RS\,(s_1+s_2) = \frac{l_{T_1}+l_{T_2}}{t_{cross}}$$

$$\frac{50}{3}+15 = \frac{100+80}{t_{cross}}$$

$$\frac{50}{3}+\frac{45}{3}$$

$$\frac{95}{3} = \frac{180}{t_{cross}}$$

$$t_{cross} = \frac{180\times 3}{95} = \frac{\cancel{180}^{36}\times 3}{\cancel{95}_{19}}$$

$$= \frac{108}{19} = 5.68 \text{ seconds}$$

18 km/hr = 5 m/s

$\downarrow \times 3$ $\downarrow \times 3$

54 km/hr = 15 m/s

18 km/hr = 5 m/s

180 km/hr = 50 m/s

$\frac{180}{3}$ km/hr = $\frac{50}{3}$ m/s

60 km/hr = $\frac{50}{3}$ m/s

18 km/hr = 5 m/s

1 km/hr = $\frac{5}{18}$ m/s

60 km/hr = $\frac{5}{\cancel{18}_{\cancel{9}\,3}}\times\frac{\cancel{60}^{\cancel{30}\,10}}{1} = \frac{50}{3}$

CONVENTIONS FOR SOLVING PROBLEM ON TRAINS

1. When a train passes an object of negligible length (Example: pole, man, stone etc), it should travel a distance equal to its own length to cross the object.

2. When a train passes a bridge or a platform or a tunnel of given length, it should travel a distance equal to the sum of its own length and the length of the bridge/platform for crossing it.

3. When two trains are moving parallel on tracks, in opposite direction => They cross each other with a relative speed equal to sum of their speeds.

4. When two trains are moving parallel on tracks, in same direction => They cross each other with a relative speed equal to difference of their speeds.

KG